

Evaluating the Jacobian Matrix

$$J = \begin{bmatrix} \frac{\partial z_0}{\partial z_0} & \frac{\partial z_0}{\partial x_1'} & \dots & \frac{\partial z_0}{\partial x_n'} \\ \vdots & \vdots & & \vdots \\ \frac{\partial z_n}{\partial z_0} & \frac{\partial z_n}{\partial x_1'} & \dots & \frac{\partial z_n}{\partial x_n'} \end{bmatrix}$$

$$\frac{\partial z_0}{\partial z_0} = 1 \quad ; \quad \frac{\partial z_0}{\partial x_j'} = 0 \quad \text{for all } j = 1, \dots, n.$$

$$\frac{\partial z_1}{\partial x_j'} = \begin{cases} 1 & \text{for } j = 1 \\ 0 & \text{otherwise.} \end{cases}$$

$$\frac{\partial z_1}{\partial z_0} = -\theta$$

$$\frac{\partial z_2}{\partial z_0} = (-\theta)^2$$

$$\text{In general } \frac{\partial z_k}{\partial z_0} = (-\theta)^k \quad k = 0, \dots, n.$$

$$\frac{\partial z_k}{\partial x_j'} = \begin{cases} (-\theta)^{k-j} & j = 1, \dots, k \\ 0 & \text{otherwise.} \end{cases}$$